2026-03-25 19:11:01 You Asked

L20.pdf

Example Lecture Scribe (structure only):

Lecture 0: Introduction to Logic

1. Definitions

Proposition: A declarative statement that is either true or false, but not both.

Notation: Denoted by lowercase letters p, q, r.

Compound Proposition: A proposition formed from existing propositions using logical operators.

2. Logical Operators

Negation ($\neg p$): The statement "it is not the case that p".

Truth Table: If p is True, $\neg p$ is False.

3. Theorems and Proofs

Theorem 1 (De Morgan's Law): $\neg(p \land q) \equiv \neg p \lor \neg q$

Proof:

Construct a truth table for both sides.

Observe that for all truth value assignments of p and q, the column for $\neg(p \land q)$ is identical to $\neg p \lor \neg q$.

Thus, the statements are logically equivalent.

4. Worked Example

Problem: Show that $p \lor \neg p$ is a tautology.

Solution:

Step 1: Assign logical values.

Step 2: If $p=T$, then $\neg p=F$, so $T \lor F = T$.

Step 3: If $p=F$, then $\neg p=T$, so $F \lor T = T$.

Conclusion: In all cases, the result is True.

Instructions:

Using the same structure as the example above, generate a lecture scribe for the provided context (lecture slides/textbook).

Requirements:

Purpose: The scribe must serve as exam reference material.

Content: Include definitions, notation, assumptions, theorem statements, proofs, derivations, and worked examples exactly as presented in the lecture materials.

Faithfulness: Do not introduce new examples, creative explanations, or intuition not discussed in the context. Everything must be traceable to the provided files.

Format: Use clear Markdown headers, bullet points for definitions, and LaTeX formatting for all mathematical notation.

Gemini 3.1 Pro

Here is the lecture scribe based on the provided materials.

Lecture 20: Linear Transformations and Expectations of Multiple RVs

1. Definitions and Statistical Representations

- [cite_start]**Random Vector**: For random variables $X_1, X_2, \ldots, X_N$, they are represented as a vector denoted by $\underline{X}$[cite: 21, 22]. [cite_start]This provides a compact representation of multiple random variables, structured as $\underline{X} = [X_1, X_2, \ldots, X_N]^T$[cite: 23, 24].
- [cite_start]**Mean Vector ($\underline{\mu}_X$)**: The expected value of a random vector, defined as $\underline{\mu}_X = E[\underline{X}]$ [cite: 31, 32]. [cite_start]It represents the average value of the random vector, where each element corresponds to the mean of one individual random variable[cite: 33, 34].
- [cite_start]**Correlation Matrix (R_{XX})**: Defined as $R_{XX} = E[\underline{X}\,\underline{X}^T]$[cite: 55]. [cite_start]Each element within the matrix is calculated as $R_{XX}(i,j) = E[X_i X_j]$[cite: 56]. [cite_start]R_{XX} is a symmetric matrix[cite: 67].
- [cite_start]**Covariance Matrix (C_{XX})**: Acts as the "relationship map" of random vectors[cite: 72]. [cite_start]The mathematical definition is $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$[cite: 74, 75]. [cite_start]The matrix is symmetric, meaning $Cov(X_i, X_j) = Cov(X_j, X_i)$[cite: 86].
- [cite_start]**Relationship between Covariance and Correlation**: $C_{XX} = R_{XX} - \underline{\mu}_X \underline{\mu}_X^T$[cite: 293].

---

2. Linear Transformations

* [cite_start]**The Linear Model**: Consider the transformation $\underline{Y} = \underline{AX} + \underline{b}$[cite: 92, 93].
    * [cite_start]$\underline{A}$ is the transformation matrix (scales and rotates the data)[cite: 94, 95].
    * [cite_start]$\underline{b}$ is the constant vector (shifts/translates the center)[cite: 96, 97].
    * [cite_start]$\underline{X}$ is the input random vector[cite: 98].
* [cite_start]**Transformation Properties**: If $\underline{Y} = \underline{AX} + \underline{b}$, the mapping properties are[cite: 203]:
    * [cite_start]**Mean**: $\mu_Y = \underline{A}\mu_X + \underline{b}$[cite: 204].
    * [cite_start]**Covariance**: $C_{YY} = \underline{A}C_{XX}\underline{A}^T$[cite: 204].
    * [cite_start]**Correlation**: $R_{YY} = \underline{A}R_{XX}\underline{A}^T + \underline{A}\mu_X\underline{b}^T + \underline{b}\mu_X^T\underline{A}^T + \underline{bb}^T$[cite: 204].

---

3. Multivariate Gaussian Distributions

* [cite_start]**1D Univariate Gaussian**: For a single RV X, the PDF is $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}}exp(-\frac{(x-\mu)^2}{2\sigma^2})$[cite: 301, 302].
* [cite_start]**N-dimensional Multivariate Gaussian**: For a vector $\underline{X} = [X_1, X_2, \dots, X_N]^T$, the general vector PDF is $f_{\underline{X}}(x) = \frac{1}{\sqrt{(2\pi)^N det(C_{XX})}}exp(-\frac{1}{2}(x - \underline{\mu}_X)^T C_{XX}^{-1}(x - \underline{\mu}_X))$[cite: 321, 323, 324].
    * [cite_start]$\underline{\mu}_X$ is the mean vector ($N \times 1$)[cite: 327].
    * [cite_start]C_{XX} is the covariance matrix ($N \times N$)[cite: 327].

---

4. Derivations and Proofs

Derivation 1: Mean of Transformed Vector

* [cite_start]**Step 1**: Start with the definition $\mu_Y = E[\underline{Y}]$[cite: 116].
* [cite_start]**Step 2**: Substitute the linear equation into the expectation: $E[\underline{AX} + \underline{b}]$[cite: 117, 119].
* [cite_start]**Step 3**: Apply linearity of expectation to separate terms: $E[\underline{AX}] + E[\underline{b}]$[cite: 129].
* [cite_start]**Result**: $\mu_Y = \underline{A}\mu_X + \underline{b}$[cite: 132].

Derivation 2: Covariance Matrix Transformation

* [cite_start]**Step 1**: Observe that the deviation from the mean is independent of the shift $\underline{b}$: $\underline{Y} - \mu_Y = (\underline{AX} + \underline{b}) - (\underline{A}\mu_X + \underline{b}) = \underline{A}(\underline{X} - \mu_X)$[cite: 191, 192].
* [cite_start]**Step 2**: Substitute this into the definition of covariance: $C_{YY} = E[(\underline{Y} - \mu_Y)(\underline{Y} - \mu_Y)^T]$[cite: 194].
* [cite_start]**Step 3**: Expand the terms: $C_{YY} = E[\underline{A}(\underline{X} - \mu_X)(\underline{X} - \mu_X)^T\underline{A}^T]$[cite: 195].
* [cite_start]**Result**: Simplified to $C_{YY} = \underline{A}C_{XX}\underline{A}^T$[cite: 197].

Proof: Uncorrelated Gaussian Random Variables

- [cite_start]**Statement**: The joint Gaussian PDF for a vector of uncorrelated random variables factorizes into the product of individual Gaussian PDFs[cite: 417, 421].
- [cite_start]**Step 1**: For mutually uncorrelated Gaussian random variables, $Cov(X_i, X_j) = 0$ for all $i \neq j$[cite: 427, 428].
- [cite_start]**Step 2**: The covariance matrix C_{XX} simplifies to a diagonal matrix, meaning its inverse C_{XX}^{-1} is also diagonal with elements σ_n^{-2}[cite: 437, 438].
- [cite_start]**Step 3**: Substituting the inverse back into the density function exponent yields $exp(-\frac{1}{2}\sum_{n=1}^{N}(\frac{x_n-\mu_n}{\sigma_n})^2)$[cite: 449, 451].
- [cite_start]**Result**: The joint PDF factorizes into $\prod_{n=1}^{N}\frac{1}{\sqrt{2\pi\sigma_n^2}}exp(-\frac{(x_n-\mu_n)^2}{2\sigma_n^2})$[cite: 419, 421].

---

5. Worked Examples

Example 1: Numerical Linear Transformation

- **Problem**: Given $\underline{Y} = \underline{AX}$ where $\underline{\mu}_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$, and $R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$. [cite_start]Calculate the mean vector, correlation matrix, and covariance matrix of $\underline{Y}$[cite: 209, 210, 211, 214, 216, 219].
- [cite_start]**Solution (Mean Vector)**: Using $\underline{\mu}_Y = \underline{A}\underline{\mu}_X$, we multiply the matrices to get $\underline{\mu}_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$[cite: 230, 233].
- **Solution (Correlation Matrix)**: Using $R_{YY} = \underline{A}R_{XX}\underline{A}^T$. [cite_start]First, compute $\underline{A}R_{XX} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix}$[cite: 242, 243]. [cite_start]Next, multiply by $\underline{A}^T$ to get $R_{YY} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$[cite: 250, 261].
- [cite_start]**Solution (Covariance Matrix)**: Using $C_{YY} = R_{YY} - \underline{\mu}_Y\underline{\mu}_Y^T$[cite: 265].

  [cite_start]Find the outer product $\underline{\mu}_Y\underline{\mu}_Y^T = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$[cite: 274, 275].

  [cite_start]Subtract this from R_{YY} to find $C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$[cite: 283, 284].

Example 2: Smart City Traffic Case Study

- [cite_start]**Scenario**: Designing a system in Ahmedabad predicting traffic congestion using transportation and climate variables[cite: 501].
- [cite_start]**Definition**: Define a random vector $X = [X_1, X_2, X_3, X_4, X_5]^T$, representing Traffic flow, Average vehicle speed, Rainfall intensity, Temperature, and Air pollution index (AQI) [cite: 503, 504, 506, 507, 508, 509, 510].
- [cite_start]**Transformation**: The city defines a Congestion Index as a linear transformation $Y = 0.6X_1 - 0.4X_2 + 0.8X_3$[cite: 566, 567].
- [cite_start]**Inference**: Assuming $X \sim \mathcal{N}(\mu, \Sigma)$, because $Y = a^T X$ is a linear transformation, Y is also Gaussian[cite: 570, 571, 573, 575].

Would you like me to elaborate on the derivation for the 2D Joint Gaussian PDF found in the slides, or is there another section you'd like to dive deeper into?